Gated Memory Networks for Short Text Classification

Abstract. Short texts such as tweets and search queries carry little context, which makes them hard to classify. We propose Gated Memory Networks (GMN), a model that stores word representations in an external memory and uses a learned gate to decide how much of the memory to read at each step. On three benchmark datasets GMN improves accuracy by 2.1 points on average over strong recurrent baselines while using 40 percent fewer parameters.

1 Introduction. Recurrent neural networks read a sentence one word at a time and compress it into a single hidden vector. For short texts this compression loses rare but decisive words. A memory network instead keeps every word vector accessible, so the classifier can attend directly to the words that matter. However, plain memory networks read the whole memory at every step, which adds noise when most words are irrelevant.

2 Method. The gate is a sigmoid unit computed from the current query and the memory summary. When the gate value is close to zero the model ignores the memory and relies on the recurrent state. We train the whole network end to end with cross entropy loss and dropout of 0.3.

3 Results. GMN reaches 88.4 percent accuracy on the TREC question dataset, compared with 86.1 percent for the best recurrent baseline. Ablation shows that removing the gate costs 1.7 points, which confirms that selective reading is the key ingredient.

4 Limitations. The memory grows linearly with the text length, so GMN is not suited to long documents. We also only evaluate on English datasets.